

- delci, dovolj daleč razpršeni, imajo $\epsilon_i, \vec{p}_i$

$$\rightarrow p_i = (\epsilon_i/c, \vec{p}_i)$$

skupni četverec
(ϵ in GK)

$$\underline{P} = \sum_i p_i = \left(\sum_i \frac{\epsilon_i}{c}, \sum_i \vec{p}_i \right) = \left(\frac{\epsilon}{c}, \vec{P} \right)$$

$\rightarrow$ če na sist. ne deluje zunanja sila, se četverec $\underline{P}$ ohrani:

$$\underline{P}_{\text{zač}} = \underline{P}_{\text{kon}} \rightarrow \text{samo ohranitev } \epsilon \text{ in } GK \rightarrow \text{ISTEM sistemu}$$

$\downarrow$ zapisano kompaktneje

Težiščni sistem (CMS - central momentum system)
(framo)

$$\underline{P}_{\text{cm}} = 0 \text{ ni dobra def., ker } LT(0) = 0$$

$\rightarrow$ na 0 postavimo samo krajevni del četverca $\underline{P}$

$$\Rightarrow \text{definiramo } \sum_i \vec{p}_i^* = 0 \quad * = \text{CM} = \text{tež.}$$

(poenostavitev: gledamo samo gibanje vzdolž x)

$$\rightarrow \boxed{\sum_i p_{ix}^* = 0}$$

v^* :

$$\sum_i p_{ix}^* = \gamma^* \left(\sum_i p_{ix} - \beta^* \sum_i \frac{\epsilon_i}{c} \right) = 0$$

$$\rightarrow \beta^* = \frac{\sum_i p_{ix} c}{\sum_i \epsilon_i} \rightarrow \boxed{v^* = \beta^* c = \frac{\sum_i p_{ix}}{\sum_i \epsilon_i} c^2}$$

invariante!

$\downarrow$ masa sistema (invariantna masa)

$$\underline{P} \cdot \underline{P} = \left(\sum_i \epsilon_i \right)^2 - \left(\sum_i \vec{p}_i c \right)^2 = \left(\sum_i \epsilon_i^* \right)^2 = (M c^2)^2$$

→ to je zelo fleksibilna formula:

$$\left(\sum_i \epsilon_i\right)_{\text{uč}}^2 - \left(\sum_i \vec{p}_i c\right)_{\text{uč}}^2 = \left(\sum_j \epsilon_j^*\right)_{\text{kon}}^2$$

→ i in j nista nujno enaka

Primer: prožni trk

- identična delca, masa m

- 1 trči v 2, ki miruje
↓
hitrost v

→ ohranitev E : $\epsilon_1 + \epsilon_2 = \epsilon_1' + \epsilon_2'$

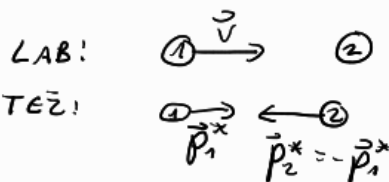
→ ohranitev GK : $\vec{p}_1 + \vec{p}_2 = \vec{p}_1' + \vec{p}_2'$

v tež. sist.:

$$\beta^* = \frac{v^*}{c} = \frac{\sum_i p_i c}{\sum_i \epsilon_i} = \frac{\cancel{\gamma m v c} + 0}{\cancel{\gamma m c^2} + m c^2} = \frac{\beta \gamma}{\gamma + 1} = \frac{\gamma - 1}{\gamma + 1}$$

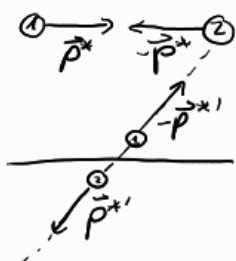
1. 2. delca

$$\gamma^* = \frac{1}{\sqrt{1 - \beta^{*2}}} = \sqrt{\frac{\gamma + 1}{2}}$$



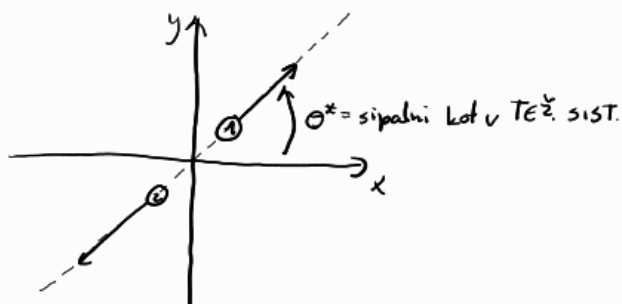
LT za GK:
$$p_2^* = \gamma^* \left(p_2 - \beta^* \frac{\epsilon_2}{c} \right) = \gamma^* \left(0 - \beta^* \frac{m c^2}{c} \right) = -\sqrt{\frac{\gamma - 1}{2}} \cdot \frac{\gamma - 1}{\gamma + 1} \cdot m c = -\sqrt{\frac{\gamma - 1}{2}} m c = -p_1^*$$

v prečnih smereh se pred trkom ne zgodi nič:



$$\epsilon_1^* = \epsilon_2^* = \epsilon_1^{*'} = \epsilon_2^{*'} = \gamma^* \left(\epsilon_2 - \beta^* p_2 c \right) = m c^2 \sqrt{\frac{\gamma + 1}{2}}$$

stanje po trku (TEŽ):



$$\left. \begin{aligned} p_{1x}^{*'} &= p_1^* \cos \theta^* \\ p_{2x}^{*'} &= -p_{1x}^{*'} = -p_1^* \cos \theta^* \\ p_{1y}^{*'} &= p_1^* \sin \theta^* \\ p_{2y}^{*'} &= -p_{1y}^{*'} = -p_1^* \sin \theta^* \end{aligned} \right\} \text{preselimo v LAB}$$

$$p'_{1x} = \gamma^* (p_{1x}^* + \beta^* \frac{E_1^*}{c}) = mc \sqrt{\frac{\gamma+1}{2}} \left(\sqrt{\frac{\gamma-1}{2}} \cos \theta^* + \sqrt{\frac{\gamma-1}{\gamma+1}} \sqrt{\frac{\gamma+1}{2}} \right) = \frac{mc}{2} \sqrt{\gamma^2-1} (1 + \cos \theta^*)$$

$$p'_{2x} = \dots = \frac{mc}{2} \sqrt{\gamma^2-1} (1 - \cos \theta^*)$$

$$p'_{1y} = \dots = \frac{mc}{2} \sqrt{\frac{\gamma-1}{2}} \sin \theta^*$$

$$p'_{2y} = \dots = -\frac{mc}{2} \sqrt{\frac{\gamma-1}{2}} \sin \theta^*$$

Limita: trk je centralen ($\theta^* = \pi$)

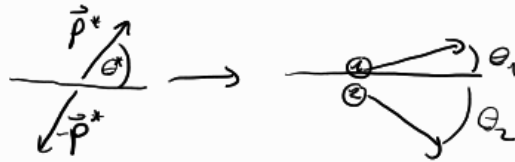
$\rightarrow$ 1 dominuje, 2 gre naprej skitir. v

$$p'_2 = p_1 \quad p'_1 = p_2 = 0$$

$\rightarrow$ kota $\theta'_1, \theta'_2 = ?$

$$\tan \theta'_1 = \frac{p'_{1y}}{p'_{1x}} = \sqrt{\frac{2}{\gamma+1}} \frac{\sin \theta^*}{1 + \cos \theta^*}$$

$$\tan \theta'_2 = \frac{p'_{2y}}{p'_{2x}} = -\sqrt{\frac{2}{\gamma+1}} \frac{\sin \theta^*}{1 - \cos \theta^*}$$



kot med delcema:

$$\tan(\theta_1 - \theta_2) = \frac{\tan \theta_1 - \tan \theta_2}{1 + \tan \theta_1 \tan \theta_2} = 2 \frac{\sqrt{2(\gamma+1)}}{\gamma-1} \frac{1}{\sin \theta^*}$$

$\rightarrow$ če v majhna: $\gamma \rightarrow 1 \rightarrow \tan(\theta_1 - \theta_2) \rightarrow \infty$
 $\downarrow$
 $= \frac{\pi}{2}$

$\rightarrow$ neprožni trk

- identična delca, u
- delca se sprimetu

$$\textcircled{1} \xrightarrow{v_1} \textcircled{2}$$

$$\text{LAB: } E_1 + E_2 = E$$

$$p_1 + p_2 = p$$

$$\gamma_1 mc^2 + mc^2 = \overbrace{\gamma}^{\text{sprimek}} M c^2$$

$$\gamma_1 m v_1 + 0 = \gamma M v$$

$$p_1 = \left(\frac{E_1}{c}, \vec{p}_1 \right)$$

$$p_2 = \left(\frac{mc^2}{c}, 0 \right)$$

$$\underline{P} = \left(\frac{E_1 + mc^2}{c}, \vec{p}_1 \right) \quad (\text{LAB})$$

$$\underline{P}^* = \left(\frac{Mc^2}{c}, 0 \right) \quad (\text{TEŽ})$$

$$\underline{P} \cdot \underline{P} = \underline{P}^* \cdot \underline{P}^*$$

$$(\gamma_1 mc^2 + mc^2)^2 - (\gamma_1 m v_1 c)^2 = (M c^2)^2$$

$$\cancel{\gamma_1^2 m^2 c^4} + 2\gamma_1 m^2 c^4 + m^2 c^4 - \cancel{\gamma_1^2 m^2 v_1^2 c^2} = M^2 c^4 \rightarrow M = \sqrt{2(\gamma_1 + 1)} m \geq 2m$$

- del mase 1 in 2 se pretvori v T , del pa v maso sprimke

$$\gamma = \frac{(\gamma_1 + 1)mc^2}{Mc^2} = \sqrt{\frac{\gamma_1 + 1}{2}} \quad \text{kin. en.}$$

deležno pri trkih $\textcircled{1} \rightarrow \textcircled{2}$ (LAB) se delca ne sprime, temveč nastanejo novi delci
 ali: $\textcircled{1} \rightarrow \textcircled{2}$ (LAB)

→ koliko česa sme nastati, pove razpoložljiva energija $E_r = m_r c^2$, $m_r = M - 2m = 2m \left(\sqrt{\frac{\gamma_1 + 1}{2}} - 1 \right)$

$$\rightarrow E_r = 2mc^2 \left(\sqrt{1 + \frac{T}{2mc^2}} - 1 \right)$$

γ_1 vsebuje v_1 , ki ustrez
 kin. en. $T = mc^2(\gamma_1 - 1)$

→ majhne T ($T \ll mc^2$): $E_r \approx \frac{T}{2}$

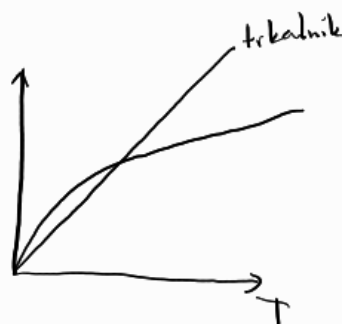
→ velike T ($T \gg mc^2$): $E_r \approx \sqrt{2mc^2 T} \propto \sqrt{T}$

⇒ - Eksperiment s fiksno tarčo: (npr. Jefferson Lab) E_r

$$\textcircled{1} \xrightarrow{v, T} \textcircled{2} : E_r \propto \sqrt{T}$$

- Trkalnik: (npr. CERN)

$$\textcircled{1} \xrightarrow{T} \textcircled{2} : E_r = 2T$$



Primer: $e^- + e^- \rightarrow e^- + e^- + e^- + e^+$
 ↑
 pozitron

$T_{\min}$ e^- , ki se zaleti v mirujoči e^- , da nastane še par $e^+ e^-$?

$$E_r = 2m_e c^2 = 2m_e c^2 \left(\sqrt{1 + \frac{T}{2m_e c^2}} - 1 \right) \rightarrow T = 6m_e c^2$$

$$\hookrightarrow m_{e^+} = m_{e^-} = 0.511 \text{ MeV}/c^2$$

zagotoviti
 moramo vsaj
 njuni mirni E

ne samo $2m_e c^2$, ker moramo dvigniti 6K!

→ γ + (delec z maso M) $\rightarrow$ (delec z maso M) + 2 · (delec z maso m)
 foton

$E_{\min} \gamma, \dots$?

Energija praga

$$(E_\gamma + Mc^2)^2 - p_\gamma^2 c^2 = (Mc^2 + 2mc^2)^2$$

$$E_\gamma^2 + 2E_\gamma Mc^2 + M^2 c^4 - E_\gamma^2 = M^2 c^4 + 4Mm c^4 + 4m^2 c^4$$

$$E_\gamma = 2mc^2 \left(1 + \frac{m}{M} \right)$$

→ če $M \rightarrow \infty$: $E_\gamma = 2mc^2$
 (tarča se ne premika)

